Revision_Deflation — the 16-component Dirac field and its singlet superfluid on spacetimes whose coordinates x5, x6, x7 deflate exponentially

PROJECT spacetimes: where to start, what was found, and what is open

Prepared for Patrick Lee Nash, Ph.D.

November 12, 2026

What this folder is

The main project (the folder above this one) studies a 16-component Dirac field on two flat eight-dimensional spacetimes, of signatures 4+4 and 7+1, and a superfluid of fermion pairs bound by a scalar field, the Vorton-Scalar. On November 12, 2026 the author paused the main project and instructed that every calculation be repeated under new hypotheses:

> the coordinates $x5$, $x6$, $x7$ deflate exponentially; Minkowski space is replaced by a pseudo-Riemannian space with a diagonal metric whose components along these coordinates are $g_{55} = -\text{const}^2 e^{-2\,\text{g55}\,x4}$, and likewise g_{66} with the rate g66 and g_{77} with the rate g77.

Here g55, g66, g77 are the author's names for three positive rates, and g_{55}, g_{66}, g_{77} are components of the metric. This folder contains the repetition: programs, data, figures, documents and two manuscripts. Nothing in the folder above was changed, except one line of a build script (recorded in the provenance P11 of the main project).

Where to start

if you want	read
the two papers	`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime.pdf` and `... on 7+1 Deflating Spacetime.pdf`
the geometry and the Dirac equations, derived step by step	`dirac matrices deflating.pdf` (DMD)
the superfluid on the deflating spacetimes	`supplement/RS1_vorton_scalar_and_superfluid_deflating.pdf`
the effective action along the deflation	`supplement/RS4_effective_action_along_the_clock.pdf`
the Higgs hypothesis	`supplement/RS2_higgs_hypothesis_deflating.pdf`
cosmology, dark matter, the Hubble tension	`supplement/RS3_cosmology_deflating.pdf`
what was assumed, what mirrors what, what is open	`provenance/RP0_hypotheses_and_status.pdf`
the proof that no symbol has two meanings	`provenance/RP1_master_symbols_deflation.pdf`
how to run everything	`provenance/RP8_code_deflation.pdf`
every error that was made	`provenance/RP9_error_log_deflation.pdf`

Every document exists as `.md`, `.tex` and `.pdf`.

What was found

Geometry (DMD, Part 1). The deflating spacetimes are homogeneous, with 14, 15 or 20 symmetries (three different, two equal, three equal rates) instead of the 36 of flat spacetime. They are not vacuum solutions of Einstein's equations. *With three equal rates they are exact vacuum solutions of Einstein–Gauss–Bonnet gravity*, for a cosmological constant $3\,\mathrm{g55}^2$ and a Gauss–Bonnet coupling $-1/(4\,\mathrm{g55}^2)$.

The Dirac field (DMD, Parts 2 to 4). A field that does not depend on $x5, x6, x7$ obeys, after a rescaling by the square root of the deflating volume, exactly the flat equations. A field that does depend on them behaves in opposite ways on the two spacetimes. On **7+1** the deflation creates particles, with a Fermi–Dirac spectrum of the temperature rate$/2\pi$. On **4+4** every admissible mode that depends on a deflating time is a standing wave that is extinguished as the exponential of an exponential: *the deflation removes the dependence on the additional times, which the main project had to impose by hand*.

The superfluid (RS1, RS4). The model of the main project holds unchanged, with a coupling that grows: $\alpha_{VS} \propto e^{\tau_d}$, where $\tau_d = \mathrm{gsum}\, x4$ is the number of e-foldings of the deflating volume; the masses are constant. Every threshold of the main project is therefore passed at a definite time. The gap grows much faster than the coupling, $d\ln(\mathrm{gap})/d\tau_d = 1/\lambda$. Two fermions become bound at a scattering resonance. The paired state can follow the deflation only once the gap exceeds a few times gsum$/\lambda$; before that it lags and breaks pairs. A vortex line carries four fermion zero modes, two for each direction.

The Higgs hypothesis (RS2). The internal symmetry that gives the pair condensate the pattern of the Higgs vacuum survives exactly, but on the deflating 7+1 spacetime it is no longer a symmetry of the spacetime. And if the condensate were the Higgs field, the electroweak scale would follow the growing coupling; atomic clocks then limit the rate of deflation to about 10^{-6} of the Hubble constant. *The Higgs hypothesis and a deflation of any consequence today exclude each other.*

Cosmology (RS3). If gravity is the eight-dimensional Einstein theory, Newton's constant grows with the deflation, lunar laser ranging limits the rate to 3×10^{-3} of the Hubble constant, and the Hubble constant inferred from the microwave background *falls*. The Hubble tension is not relieved. For dark matter the deflation adds an epoch of resonant self-interaction and a small change of the growth rate; the conclusions of the main project about the masses are unchanged.

What is open

1. The spectra of the microwave background with a time-dependent Newton's constant were not computed (RP0, Sec. 5). The author's decision is requested.
2. The stability of the Einstein–Gauss–Bonnet solution, and the source of the deflation for unequal rates.
3. The back-reaction of the created particles (7+1), and a principle that excludes the growing solutions (4+4).
4. The end of every history: the couplings grow without bound, and the model fails when they become strong.

The stages of the main project that were paused are listed in its provenance P11 and remain paused.

Numbers

The Rust program `deflation` performs 217 checks; an independent Python script 33 of 33; the registry of symbols 6 of 6. All pass. The folder contains 14 figures and the data files from which every table and figure is made.

How to run it

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
cargo test  --release --manifest-path code/deflation-rs/Cargo.toml
for c in matrices modes history quench cosmology symmetry channels matter paired; do
  ./code/deflation-rs/target/release/deflation $c
done
python3 code/scripts/verify_deflating_independent.py
```

The provenance RP8 explains every step, and the remaining ones (symbols, tables, figures, documents).